ADVANCED MATERIALS ENGINEERING · POLYMERS AND COMPOSITES

Composites in the real world: how carbon fiber went from F1 to furniture

Joshua Otaigbe · December 15, 2026

In 1981, John Watson crashed his McLaren at Monza at roughly 170 mph. The car disintegrated. Watson climbed out and walked to the medical center.

The McLaren MP4/1 was the first Formula 1 car built with a full carbon fiber monocoque chassis, designed by John Barnard. Before it, F1 cars were made of aluminum. After it, every serious racing car was made of carbon fiber composites. The crash at Monza wasn't just a lucky escape. It was a public demonstration, watched by millions, of what carbon fiber could do.

At the time, carbon fiber cost around $150 per kilogram. It lived in fighter jets, aerospace test vehicles, and the budgets of Formula 1 teams. The idea that it would one day show up in bicycles, laptops, golf clubs, and prosthetic running blades would have seemed far-fetched. But that's exactly what happened, and the story of how is one of the cleaner examples of how advanced materials move from laboratory to consumer shelf.

The material itself was developed in the early 1960s at the Royal Aircraft Establishment in Farnborough, England, where William Watt and his colleagues discovered that heating polyacrylonitrile (PAN) fiber in a controlled atmosphere produced fibers of near-pure carbon with extraordinary stiffness. The carbon atoms arrange themselves along the fiber axis in a graphite structure. That alignment is why the fiber is so stiff: you're loading directly along the crystallographic direction of maximum stiffness, not at some diagonal to it.

But a carbon fiber alone isn't a structural material. It's a strand roughly 5 to 10 microns in diameter, finer than a human hair, with no ability to carry compression loads without buckling. Carbon fiber reinforced polymer (CFRP) works by embedding thousands of those fibers in a polymer matrix, usually an epoxy resin, which holds the fibers in position, transfers load between them and protects the fiber surface. The composite's properties depend on fiber volume fraction, fiber orientation, number of plies, and cure cycle. A skilled composite designer specifies each of these to match the load case. An aerospace spar loaded primarily in bending gets very different layup choices than a racing car tub loaded in torsion.

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Making CFRP starts with PAN precursor fiber, which goes through an oxidation stage at 200 to 300°C, then carbonization at 1,000 to 1,500°C in an inert nitrogen atmosphere, which drives off everything except carbon. What comes out is a fiber with tensile strength around 3,500 to 7,000 MPa depending on grade. Compare that to structural steel at roughly 400 to 600 MPa. The fiber is then surface treated to improve bonding with the resin matrix, coated with a sizing compound to protect it during handling, and either woven into fabric or impregnated with resin to make prepreg: pre-impregnated tape or fabric, ready to lay up and cure.

Toray Industries in Japan is the world's largest carbon fiber manufacturer. The Boeing 787 Dreamliner, which entered commercial service in 2011, uses Toray fiber throughout a fuselage that's 50% composites by weight. Airbus followed with the A350, at 53%. Before the 787, commercial aircraft fuselages were aluminum. The switch to CFRP saved roughly 20% in structural weight, which compounded into significant fuel savings over each aircraft's operating lifetime.

The cost drop that made this possible didn't come from a single invention. It came from scale. As aerospace demand grew, manufacturers invested in larger, faster production lines. Fiber precursor costs fell. Process yields improved. The 787 programme, with its enormous composite content, pulled the industry toward scale it had never needed before. Standard modulus carbon fiber, which cost around $150/kg in the 1970s, now sits around $20 to $30/kg. High-modulus aerospace grade is still expensive, but the commodity end of the market is accessible to almost any manufacturer willing to design for it.

That accessibility is why CFRP is now in road bicycles, tennis rackets, fishing rods, laptop shells, automotive body panels, and wind turbine blades. The Specialized S-Works Tarmac, a top-end road bike, uses carbon fiber that Specialized co-developed with Toray. A complete frame weighs under 700 grams. A racing driver's prosthetic running blade stores and returns elastic energy in a way no titanium or steel blade could match, because CFRP can be laid up to be stiff in exactly one direction and flexible in another, within the same part.

The recycling problem is real and largely unsolved.

Most aerospace-grade CFRP uses thermoset epoxy resins, which cure by forming a crosslinked polymer network that can't be re-melted. When a composite aircraft part reaches end of life, or when manufacturing produces scrap, there's no simple recycling route. Pyrolysis, which burns off the resin in a low-oxygen atmosphere, recovers the carbon fiber but shortens the fibers and reduces their properties. Solvolysis (chemical decomposition of the resin) recovers higher-quality fiber but is expensive and energy-intensive. Most composite waste currently goes to landfill.

Thermoplastic CFRP uses a polymer matrix that can be re-melted rather than a thermoset one.

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Thermoplastic composites can be welded, reshaped, and recycled at end of life. They're already used in short-cycle aerospace structures and are being actively developed by Airbus, Fokker, and several Tier 1 suppliers for primary structure. The recycling infrastructure still needs to be built.

Beyond CFRP, the frontier in structural composites is ceramic matrix composites. CMCs use ceramic fibers (usually silicon carbide) in a ceramic matrix, producing a material that can operate at temperatures far above what any polymer composite can tolerate. The GE LEAP engine, used in the Boeing 737 MAX and Airbus A320neo, uses SiC/SiC CMC components in its high-pressure turbine section. The CMC blades are roughly 30% lighter than the nickel superalloy components they replaced, and they allow the engine to operate at higher temperatures, which improves thermal efficiency.

Making CMC parts is significantly more complicated than making CFRP. The processes involve chemical vapor infiltration of the matrix into a woven ceramic preform, at high temperatures, over long cycle times. The material is brittle in a different way from metals, which requires a complete rethinking of design approaches and damage tolerance methodology. It's expensive. But the performance gains at high temperature are real, and the aerospace industry is the kind of buyer that can justify expensive materials when the lifetime operating cost savings come back around.

Carbon fiber went from a laboratory curiosity in Farnborough to an Olympic bicycle frame in roughly 40 years. Ceramic matrix composites are on a shorter arc, driven by the hard thermal limits of jet engine efficiency. The pattern is consistent across the history of advanced materials: cost follows scale, scale follows demand, and demand follows performance that can't be achieved any other way.

For more information or if you have any questions, please contact the author.

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